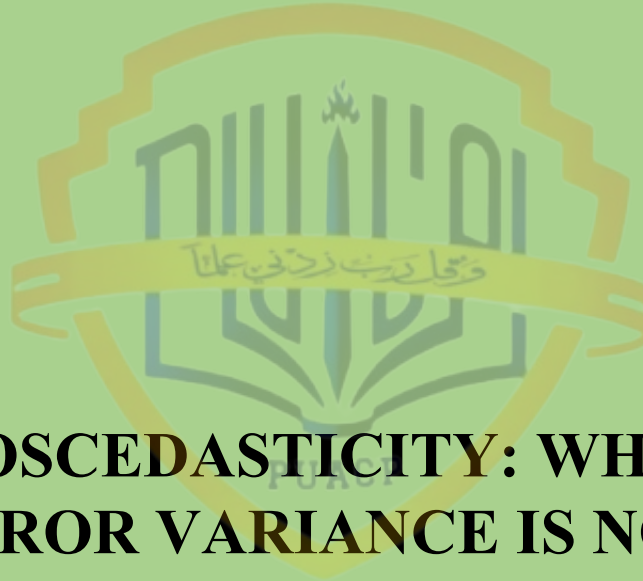




HETEROSCEDASTICITY: WHAT HAPPENS IF THE ERROR VARIANCE IS NONCONSTANT?

Damodar N. Gujarati and Dawn C Porter (2009)

Learning Objectives

We seek answers to the following questions:

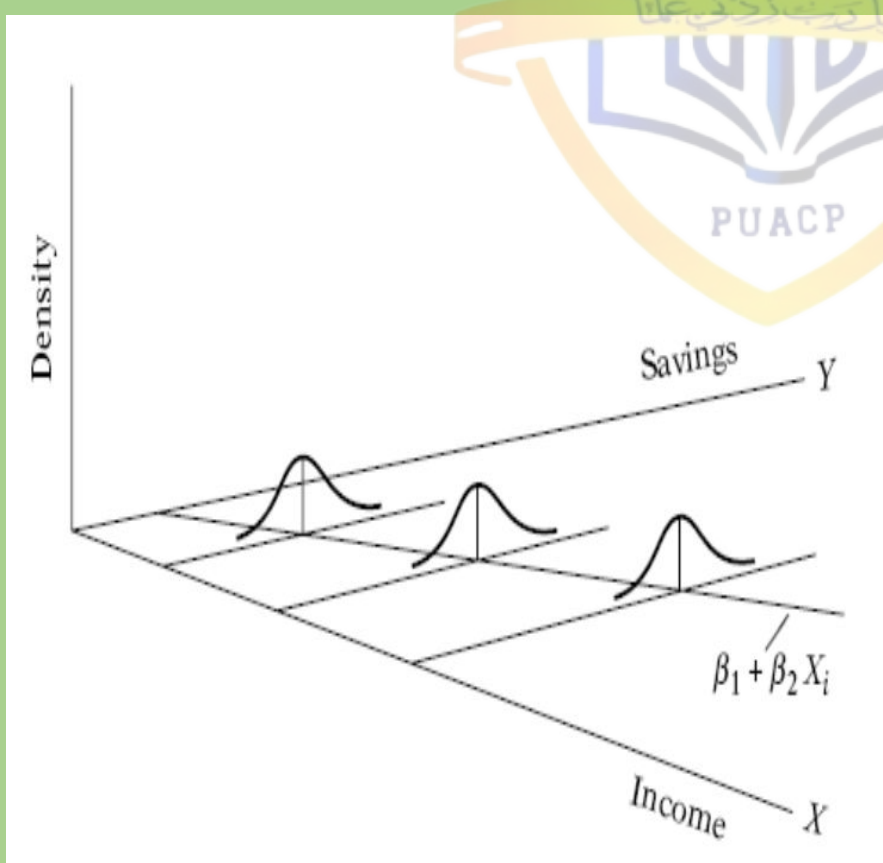
- 1. What is the nature of heteroscedasticity?
- 2. What are its consequences?
- 3. How does one detect it?
- 4. What are the remedial measures?

THE NATURE OF HETEROSCEDASTICITY

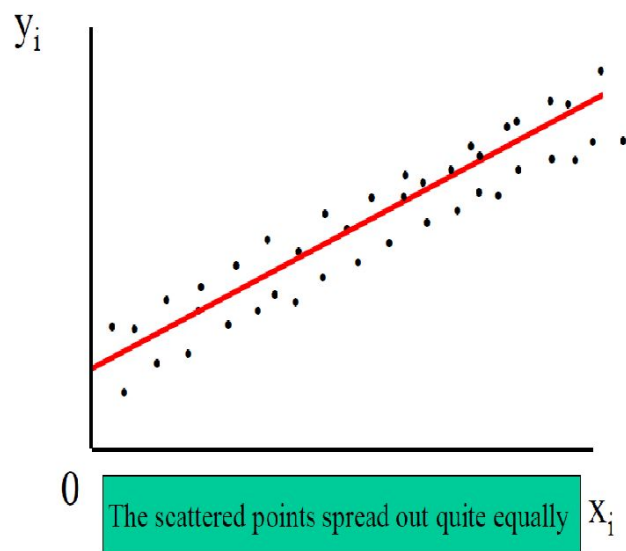
- One of the important assumptions of the classical linear regression model is that the *variance of each disturbance term u_i* , conditional on the chosen values of the explanatory variables, is some constant number equal to σ^2 .
- This is the assumption of homoscedasticity, or *equal (homo) spread (scedasticity)*, that is, *equal variance*. Symbolically,
- $E(u_i^2) = \sigma^2 \quad i = 1, 2, \dots, n \quad (1)$

Continued

Figure 1: Homoscedastic (Equal Spread)



Homoscedastic pattern of errors



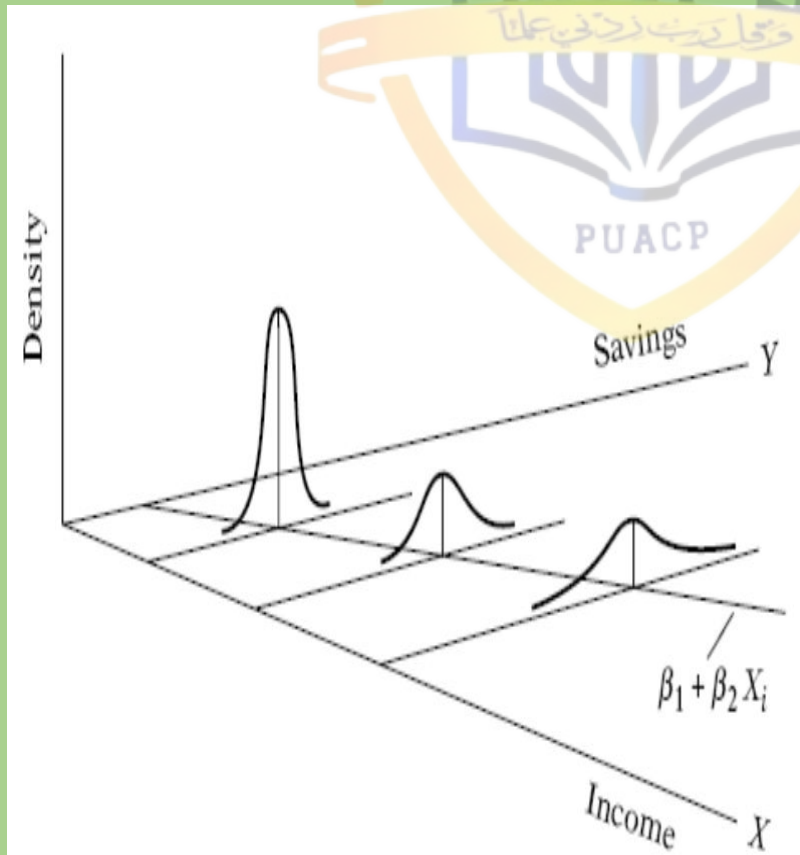
Continued

- In contrast to Figure 1, if the variances of Y_i are not the same. Hence, there is heteroscedasticity. Symbolically,

$$E(u^2_i) = \sigma^2_i \quad (2)$$

- Notice the subscript of σ^2 , which reminds us that the conditional variances of u_i (= conditional variances of Y_i) are no longer constant.

Continued



Heteroscedastic pattern of errors

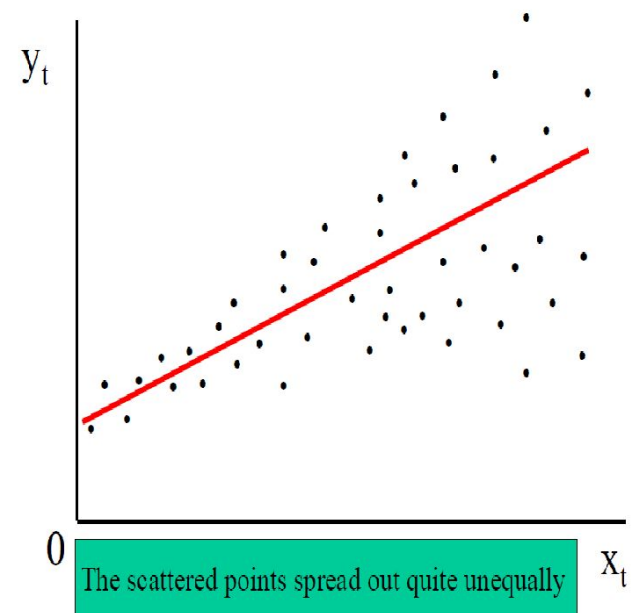


Figure 2: Heteroscedasticity (non-equal spread)

Continued

- Note that the problem of heteroscedasticity is likely to be **more common in cross-sectional data** than in time series.
- In cross-sectional data, members may be of different sizes, such as small, medium, or large firms or low, medium, or high income families or countries.
- In time series data, on the other hand, the variables tend to be of similar orders of magnitude. Examples are GNP, consumption expenditure, savings.

Difference between homoscedasticity and heteroscedasticity

- Assume that in the two-variable model $Y_i = \beta_1 + \beta_2 X_i + u_i$, Y represents savings and X represents income. Figures 1 and 2 show that as income increases, savings on the average also increase. But in Figure 1 the variance of savings remains the same at all levels of income, whereas in Figure 2 it increases with income. It seems that in Figure 2 the *higher income families on the average save more than the lower-income families*, but there is also *more variability* in their savings.

Reasons for heteroscedasticity

- There are several reasons why the variances of u_i may be variable, some of which are as follows.
- 1. Following the *error-learning models*, as people learn, their errors of behavior become smaller over time. In this case, σ_i^2 is expected to decrease. As an example, consider Figure 3, which relates the number of typing errors made in a given time period on a test to the hours put in typing practice.

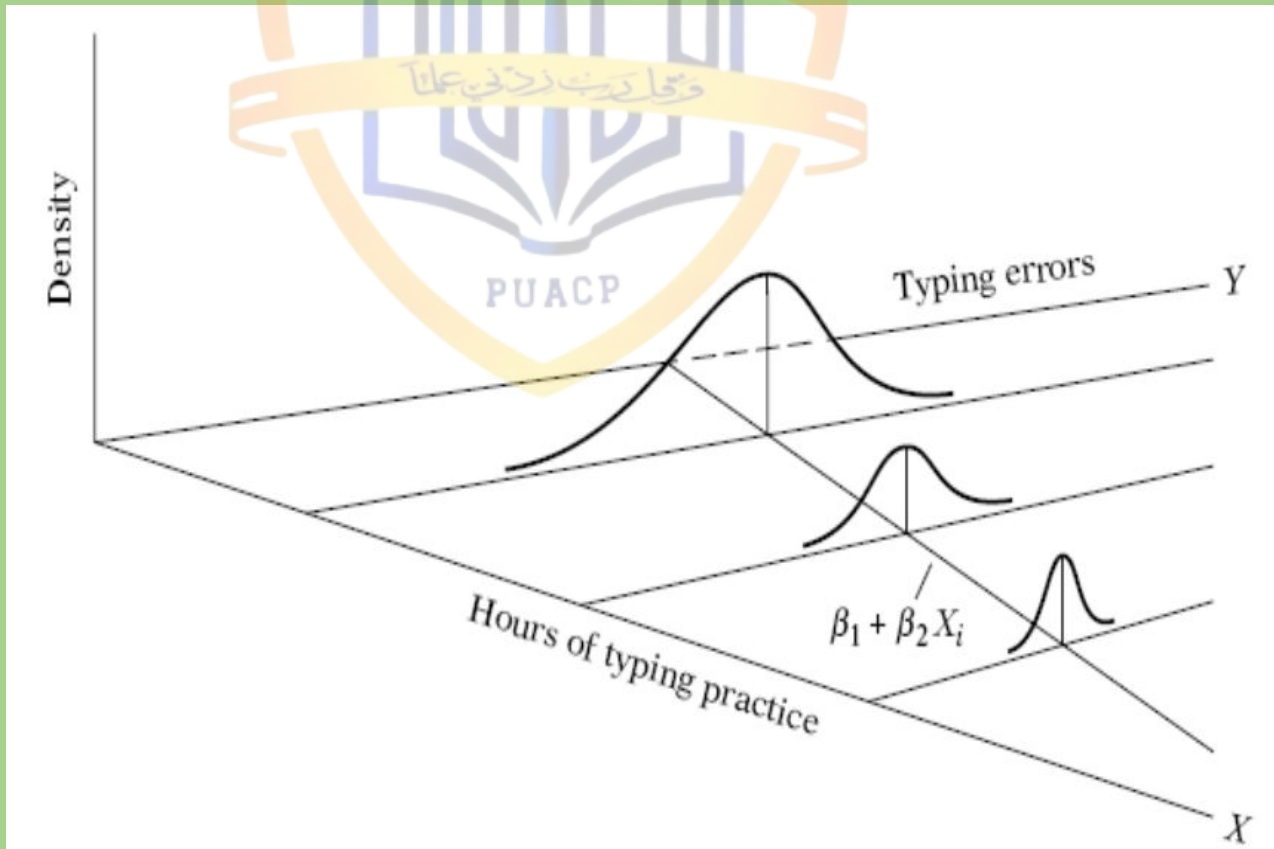


Figure 3: Illustration of heteroscedasticity

Reasons for heteroscedasticity

Continued...

- 2. As incomes grow, people have more *discretionary income* and hence more scope for choice about the disposition of their income. Hence, σ_i^2 is likely to increase with income. Similarly, companies with larger profits are generally expected to show greater variability in their dividend policies than companies with lower profits.
- 3. As data collecting techniques improve, σ_i^2 is likely to decrease. Thus, banks that have sophisticated data processing equipment are likely to commit *fewer errors* in the monthly or quarterly statements of their customers than banks without such facilities.

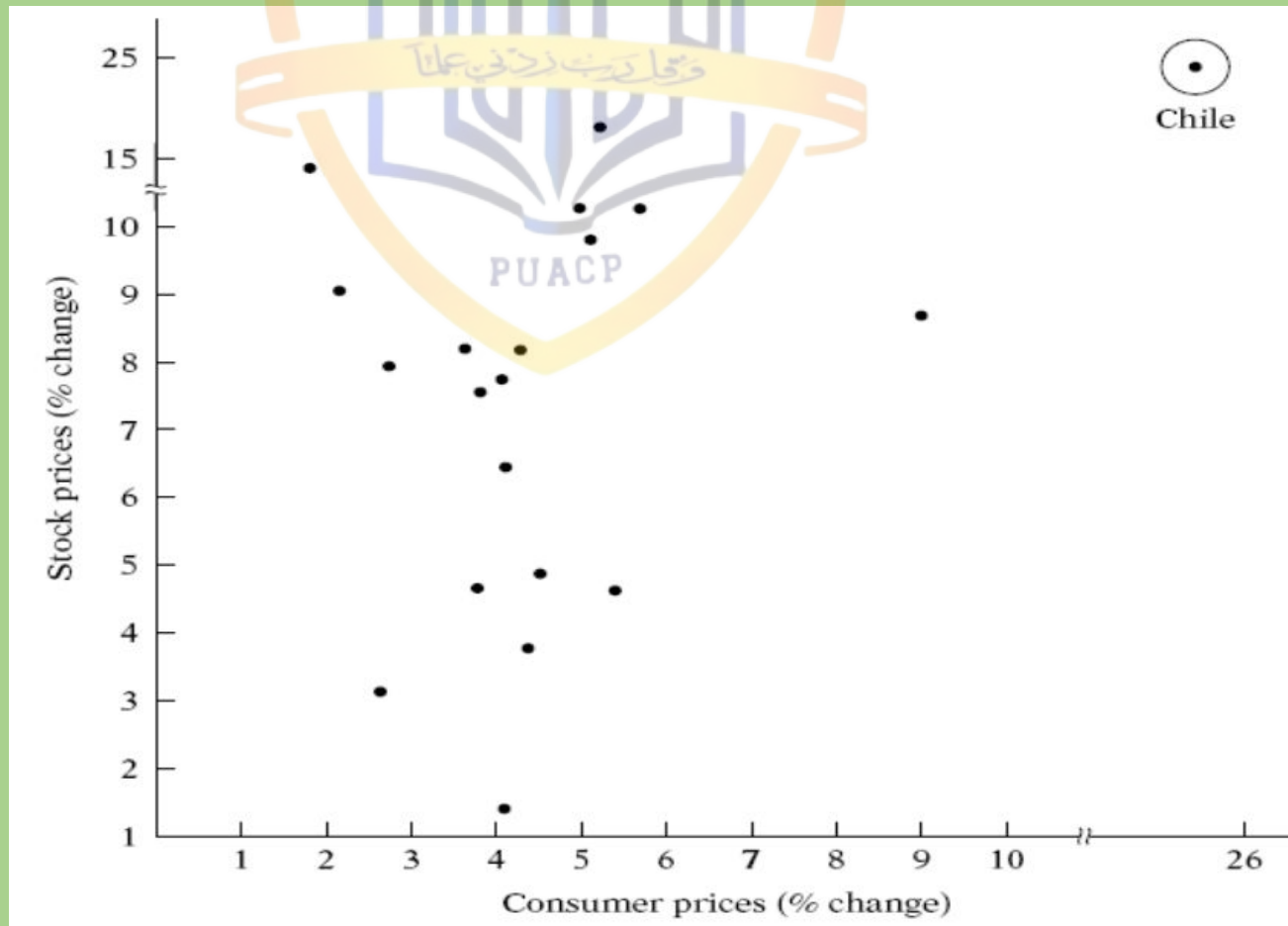
Reasons for heteroscedasticity

Continued...

- 4. Heteroscedasticity can also arise as a result of the presence of *outliers*, (either very small or very large) in relation to the observations in the sample Figure 4. The inclusion or exclusion of such an observation, especially if the sample size is small, can substantially alter the results of regression analysis. Chile can be regarded as an outlier because the given Y and X values are much larger than for the rest of the countries. In situations such as this, it would be hard to maintain the assumption of homoscedasticity.

Reasons for heteroscedasticity

Continued...



- Figure 4: The relationship between stock prices and consumer prices

Reasons for heteroscedasticity

Continued...

- 5. Another source of heteroscedasticity arises from violating Assumption 9 of CLRM, namely, that the regression model is *correctly specified*, very often what looks like heteroscedasticity may be due to the fact that some important variables are omitted from the model. But if the omitted variables are included in the model, that impression may disappear.
- 6. Another source of heteroscedasticity is *Skewness* in the distribution of one or more regressors included in the model. Examples are economic variables such as income, wealth, and education. It is well known that the distribution of income and wealth in most societies is *uneven*, with the bulk of the income and wealth being owned by a few at the top.

Reasons for heteroscedasticity

Continued...

- 7. Other sources of heteroscedasticity: As David Hendry notes, heteroscedasticity can also arise because of
 - (1) incorrect data transformation (e.g., ratio or first difference transformations)
 - (2) incorrect functional form (e.g., linear versus log–linear models).

Definition of Heteroscedasticity:

$$\text{Var}(u_i) = E(u_i^2) = \sigma_i^2 \neq \sigma^2$$

Two-variable regression: $Y = \beta_1 + \beta_2 X + u_i$

$$\hat{\beta}_2 = \frac{\sum xy}{\sum x^2} = \sum k_i Y = \sum k_i (\beta_1 + \beta_2 X + u_i)$$

$$\begin{aligned} \sum k_i &= 0 \\ \sum k_i X_i &= 1 \end{aligned}$$

$$\Rightarrow \hat{\beta}_2 = \beta_2 + \sum k_i u_i \quad E(\hat{\beta}_2) = \beta_2 \quad \text{unbiased}$$

$$\text{Var}(\hat{\beta}_2) = E(\hat{\beta}_2 - \beta_2)^2 = E(\sum k_i u_i)^2$$

$$= E(k_1^2 u_1^2 + k_2^2 u_2^2 + \dots + 2k_1 k_2 u_1 u_2 + \dots)$$

$$= k_1^2 \sigma_1^2 + k_2^2 \sigma_2^2 + \dots + 0 + \dots$$

$$= \sum k_i^2 \sigma_i^2$$

$$= \frac{\sum x_i^2 \sigma_i^2}{(\sum x_i^2)^2}$$

if $\sigma_1^2 \neq \sigma_2^2 \neq \sigma_3^2 \neq \dots$
i.e., heteroscedasticity

$$\text{Var}(\hat{\beta}_2) = \frac{\sigma^2}{\sum x_i^2}$$

if $\sigma_1^2 = \sigma_2^2 = \sigma_3^2 = \dots$
i.e., homoscedasticity

Consequences of heteroscedasticity

1. OLS estimators are still linear and unbiased

2. $\text{Var}(\hat{\beta}_1)$ s are not minimum.

=> not the best => not efficiency => not BLUE

3. $\text{Var}(\hat{\beta}_2) = \frac{\sum x_i \sigma_i^2}{(\sum x_i^2)^2}$ instead of $\text{Var}(\hat{\beta}_2) = \frac{\sigma^2}{\sum x_i^2}$

Two-variable
case

4. $\hat{\sigma}^2 = \frac{\sum \hat{u}_i^2}{n-k}$ is biased, $E(\hat{\sigma}^2) \neq \sigma^2$

5. t and F statistics are unreliable.

$$Y = \beta_0 + \beta_1 X + u$$

$$\text{SEE} = \hat{\sigma} \quad \text{RSS} = \sum \hat{u}^2$$

Cannot be min.

11.3 THE METHOD OF GENERALIZED LEAST SQUARES (GLS)

- *If u_i is heteroscedastic $\hat{\beta}_2$ is no longer best, although it is still unbiased.*

Reason: refer to Table on slide 19 if we were to regress per-employee compensation on the size of employment, we would like to make use of the knowledge that there is considerable interclass variability in earnings.

- Ideally, we would like to devise the estimating scheme in such a manner that observations coming from populations with greater variability are given less weight than those coming from populations with smaller variability.

Continued...

COMPENSATION PER EMPLOYEE (\$) IN NONDURABLE MANUFACTURING INDUSTRIES ACCORDING TO EMPLOYMENT SIZE OF ESTABLISHMENT, 1958

Industry	Employment size (average number of employees)								
	1-4	5-9	10-19	20-49	50-99	100-249	250-499	500-999	1000-2499
Food and kindred products	2994	3295	3565	3907	4189	4486	4676	4968	5342
Tobacco products	1721	2057	3336	3320	2980	2848	3072	2969	3822
Textile mill products	3600	3657	3674	3437	3340	3334	3225	3163	3168
Apparel and related products	3494	3787	3533	3215	3030	2834	2750	2967	3453
Paper and allied products	3498	3847	3913	4135	4445	4885	5132	5342	5326
Printing and publishing	3611	4206	4695	5083	5301	5269	5182	5395	5552
Chemicals and allied products	3875	4660	4930	5005	5114	5248	5630	5870	5876
Petroleum and coal products	4616	5181	5317	5337	5421	5710	6316	6455	6347
Rubber and plastic products	3538	3984	4014	4287	4221	4539	4721	4905	5481
Leather and leather products	3016	3196	3149	3317	3414	3254	3177	3346	4067
Average compensation	3396	3787	4013	4104	4146	4241	4388	4538	4843
Standard deviation	742.2	851.4	727.8	805.06	929.9	1080.6	1241.2	1307.7	1110.5
Average productivity	9355	8584	7962	8275	8389	9418	9795	10,281	11,750

Source: *The Census of Manufacturers*, U.S. Department of Commerce, 1958 (computed by author).

Continued...

- Examining Table on slide 19, we would like to weight observations coming from employment classes 10–19 and 20–49 more heavily than those coming from employment classes like 5–9 and 250–499, for the former are more closely clustered around their mean values than the latter, thereby enabling us to estimate the PRF more accurately.

Continued...

- Unfortunately, the usual OLS method does not follow this strategy, but a method of estimation, known as **generalized least squares** (GLS), takes such information into account explicitly and is therefore capable of producing estimators that are BLUE. To see how this is accomplished, let us continue with two-variable model:

- $$Y_i = \beta_1 + \beta_2 X_i + u_i \quad (3)$$

which for ease of algebraic manipulation we write as

- $$Y_i = \beta_1 X_{0i} + \beta_2 X_i + u_i \quad (4)$$

- where $X_{0i} = 1$ for each i . Now assume that the heteroscedastic variances σ_i^2 are known. Divide through by σ_i to obtain:

-

Continued...

$$\frac{Y_i}{\sigma_i} = \beta_1 \left(\frac{X_{0i}}{\sigma_i} \right) + \beta_2 \left(\frac{X_i}{\sigma_i} \right) + \beta_1 \left(\frac{u_i}{\sigma_i} \right) \quad (6)$$

which for ease is written as

$$Y^*_i = \beta^*_1 X^*_{0i} + \beta^*_2 X^*_i + u^*_i \quad (7)$$

$$\text{and } \text{var}(u_i^*) = E(u_i^*)^2 = E\left(\frac{u_i}{\sigma_i}\right)^2 = \left(\frac{1}{\sigma_i^2}\right) \cdot E(u_i^2) = \left(\frac{1}{\sigma_i^2}\right) \cdot \sigma_i^2 = 1$$

- The transformed disturbance term u_i^* is homoscedastic. If we apply OLS on the transformed model. I produce BLUE estimates.
- This procedure of transforming the original variables in such a way that the transformed variables satisfy the assumptions of the classical model and then applying OLS to them is known as *the method of generalized least squares (GLS)*.
- GLS is OLS on the transformed variables that satisfy the standard least squares assumptions.

Difference between GLS and OLS

- Recall from Chapter 3 that in OLS we minimize:

$$\sum \hat{u}_i^2 = \sum (Y_i - \hat{\beta}_1 - \hat{\beta}_2 X_i)^2$$

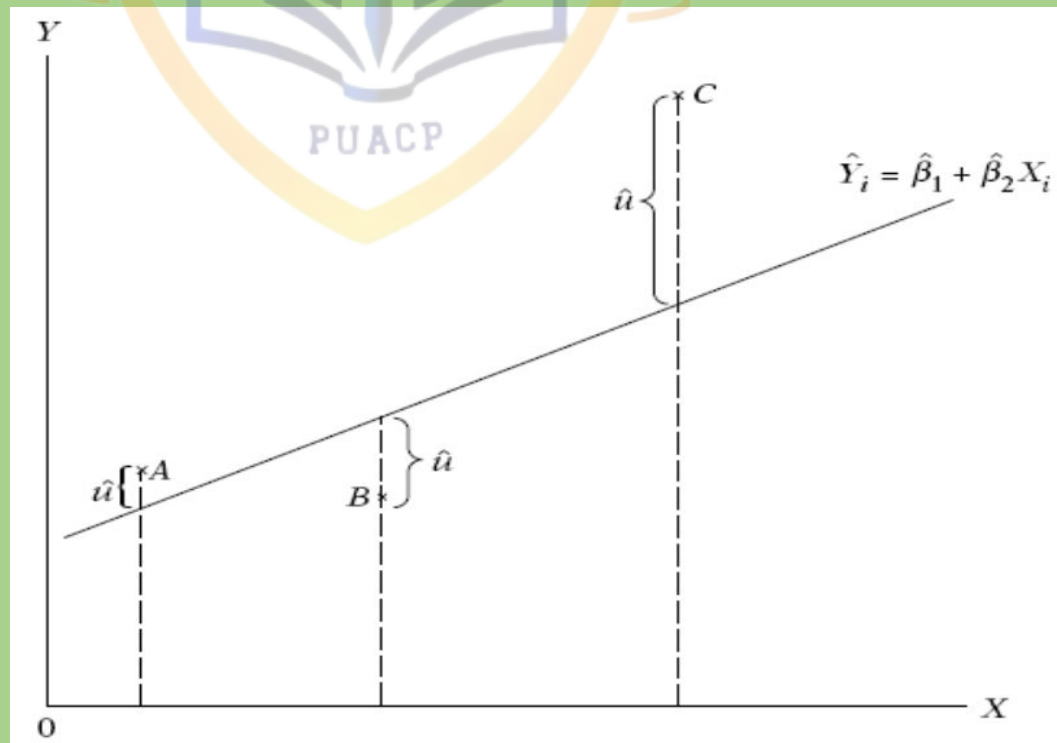
but in GLS we minimize the expression

$$\sum w_i \hat{u}_i^2 = \sum w_i (Y_i - \hat{\beta}_1^* X_{0i} - \hat{\beta}_2^* X_i)^2$$

- where $w_i = 1/\sigma_i^2$
- Thus, in GLS we minimize a weighted sum of residual squares with $w_i = 1/\sigma_i^2$ acting as the weights, but in OLS we minimize an unweighted or (what amounts to the same thing) equally weighted RSS.
- In GLS the weight assigned to each observation is inversely proportional to its σ_i , that is, observations coming from a population with larger σ_i will get relatively smaller weight and those from a population with smaller σ_i will get proportionately larger weight in minimizing the RSS (9).

Continued...

- To see the difference between OLS and GLS clearly, consider the hypothetical scattergram given in Figure 5.



- Figure 5: Hypothetical Scattergram

Continued...

- In the (unweighted) OLS, each u_i^2 associated with points A , B , and C will receive the same weight in minimizing the RSS. Obviously, in this case the u_i^2 associated with point C will dominate the RSS. But in GLS the extreme observation C will get relatively smaller weight than the other two observations.
- Since (9) minimizes a weighted RSS, it is appropriately known as weighted least squares (WLS), and the estimators thus obtained are known as WLS estimators.
- But WLS is just a special case of the more general estimating technique, GLS. In the context of heteroscedasticity, one can treat the two terms WLS and GLS interchangeably.

Consequences of Using OLS in the Presence of Heteroscedasticity

- Both $\hat{\beta}_2^*$ and $\hat{\beta}_2$ are (linear) unbiased estimators but we know that it is $\hat{\beta}_2^*$ that is efficient, that is, has the smallest variance.
- What happens to our confidence interval, hypotheses testing, and other procedures if we continue to use the OLS estimator $\hat{\beta}_2$? We distinguish two cases.

OLS Estimation Allowing for Heteroscedasticity

- Using OLS variance, and assuming σ_i^2 are known, can we establish confidence intervals and test hypotheses with the usual t and F tests?

Continued....

- The answer generally is no because it can be shown that $\text{var}(\hat{\beta}_2^*) \leq \text{var}(\hat{\beta}_2)$, which means that confidence intervals based on the latter will be *unnecessarily larger*.
- As a result, the t and F tests are likely to give us *inaccurate results*.

• OLS Estimation Disregarding Heteroscedasticity

- The situation can become serious if we not only use *OLS* estimates but also continue to use the usual (homoscedastic) variance formula given even if heteroscedasticity is present or suspected, whatever conclusions we draw or inferences we make may *be very misleading*.

Continued....

- OLS, with or without correction for heteroscedasticity, overestimates the standard errors.

Value of α	Standard error of $\hat{\beta}_1$			Standard error of $\hat{\beta}_2$		
	OLS	OLS _{het}	GLS	OLS	OLS _{het}	GLS
0.5	0.164	0.134	0.110	0.285	0.277	0.243
1.0	0.142	0.101	0.048	0.246	0.247	0.173
2.0	0.116	0.074	0.0073	0.200	0.220	0.109
3.0	0.100	0.064	0.0013	0.173	0.206	0.056
4.0	0.089	0.059	0.0003	0.154	0.195	0.017

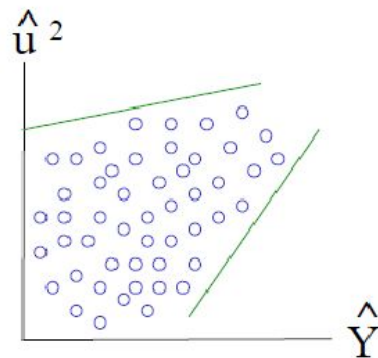
Note: OLS_{het} means OLS allowing for heteroscedasticity.

Detection of heteroscedasticity

1. Graphical method :

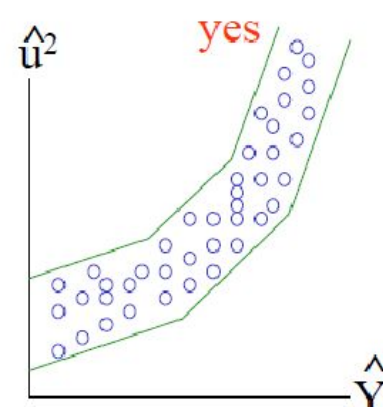
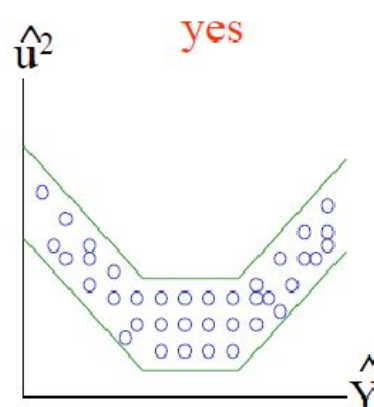
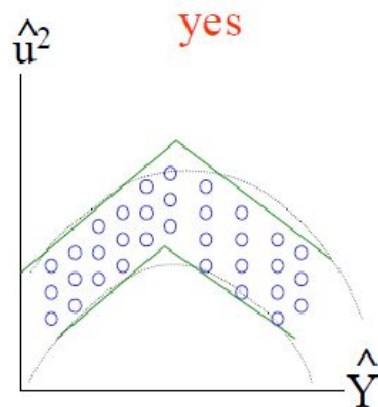
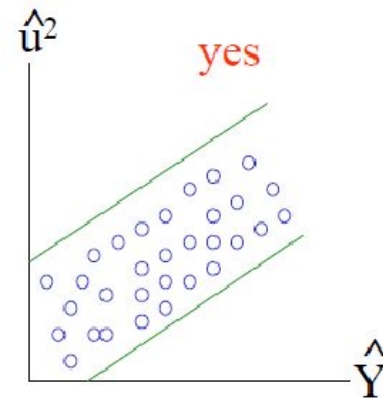
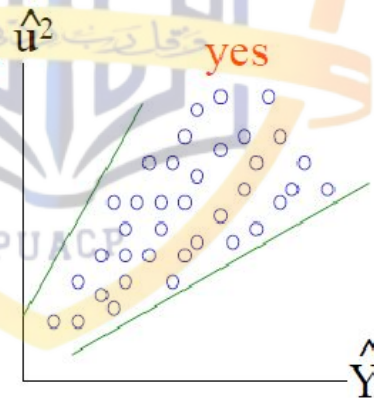
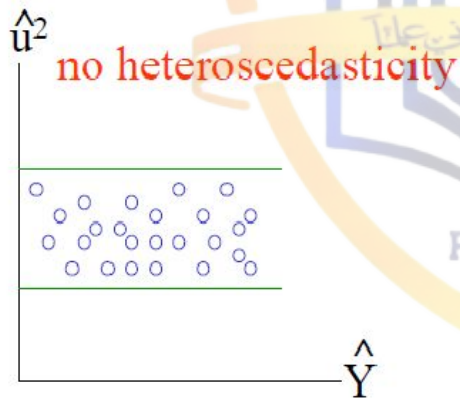
plot the estimated residual ($\hat{u}_i$) or squared ($\hat{u}_i^2$) against the predicted dependent Variable ($\hat{Y}_i$) or any independent variable (X_i).

Observe the graph whether there is a systemic pattern as:



Yes,
heteroscedasticity exists

Detection of heteroscedasticity: Graphical method



Statistical test: (i) Park test

H_0 : No heteroscedasticity exists i.e., $\text{Var}(u_i) = \sigma^2$
(homoscedasticity)

H_1 : Yes, heteroscedasticity exists i.e., $\text{Var}(u_i) = \sigma_i^2$

Park test:

Step 1. Run OLS on regression: $Y_i = \beta_1 + \beta_2 X_i + u_i$, obtain $\hat{u}_i$

Step 2. Take square and take log : $\ln(\hat{u}_i^2)$

Step 3. Run OLS on regression: $\ln(\hat{u}_i^2) = \beta_1^* + \beta_2^* \ln X_i + v_i$

Step 4. Use **t-test** to test $H_0 : \beta_2^* = 0$ (Homoscedasticity)

Suspected variable
that causes
heteroscedasticity

If $t^* > t^c \implies$ **reject H_0** $\implies$ **heteroscedasticity exists**

If $t^* < t^c \implies$ **not reject H_0** $\implies$ **homoscedasticity**

Park Test

EXAMPLE 11.1

*Relationship
between
Compensation
and Productivity*

To illustrate the Park approach, we use the data given in Table 11.1 to run the following regression:

$$Y_i = \beta_1 + \beta_2 X_i + u_i$$

where Y = average compensation in thousands of dollars, X = average productivity in thousands of dollars, and i = i th employment size of the establishment. The results of the regression are as follows:

$$\begin{aligned} \hat{Y}_i &= 1992.3452 + 0.2329X_i \\ \text{se} &= (936.4791) \quad (0.0998) \\ t &= (2.1275) \quad (2.333) \quad R^2 = 0.4375 \end{aligned} \tag{11.5.3}$$

The results reveal that the estimated slope coefficient is significant at the 5 percent level on the basis of a one-tail t test. The equation shows that as labor productivity increases by, say, a dollar, labor compensation on the average increases by about 23 cents.

The residuals obtained from regression (11.5.3) are then regressed on X_i as suggested in Eq. (11.5.2), giving the following results:

$$\begin{aligned} \widehat{\ln \hat{u}_i^2} &= 35.817 - 2.8099 \ln X_i \\ \text{se} &= (38.319) \quad (4.216) \\ t &= (0.934) \quad (-0.667) \quad R^2 = 0.0595 \end{aligned} \tag{11.5.4}$$

Obviously, there is no statistically significant relationship between the two variables. Following the Park test, one may conclude that there is no heteroscedasticity in the error variance.¹³

Example: Studenmund (2001), Equation 10.24, pp.370

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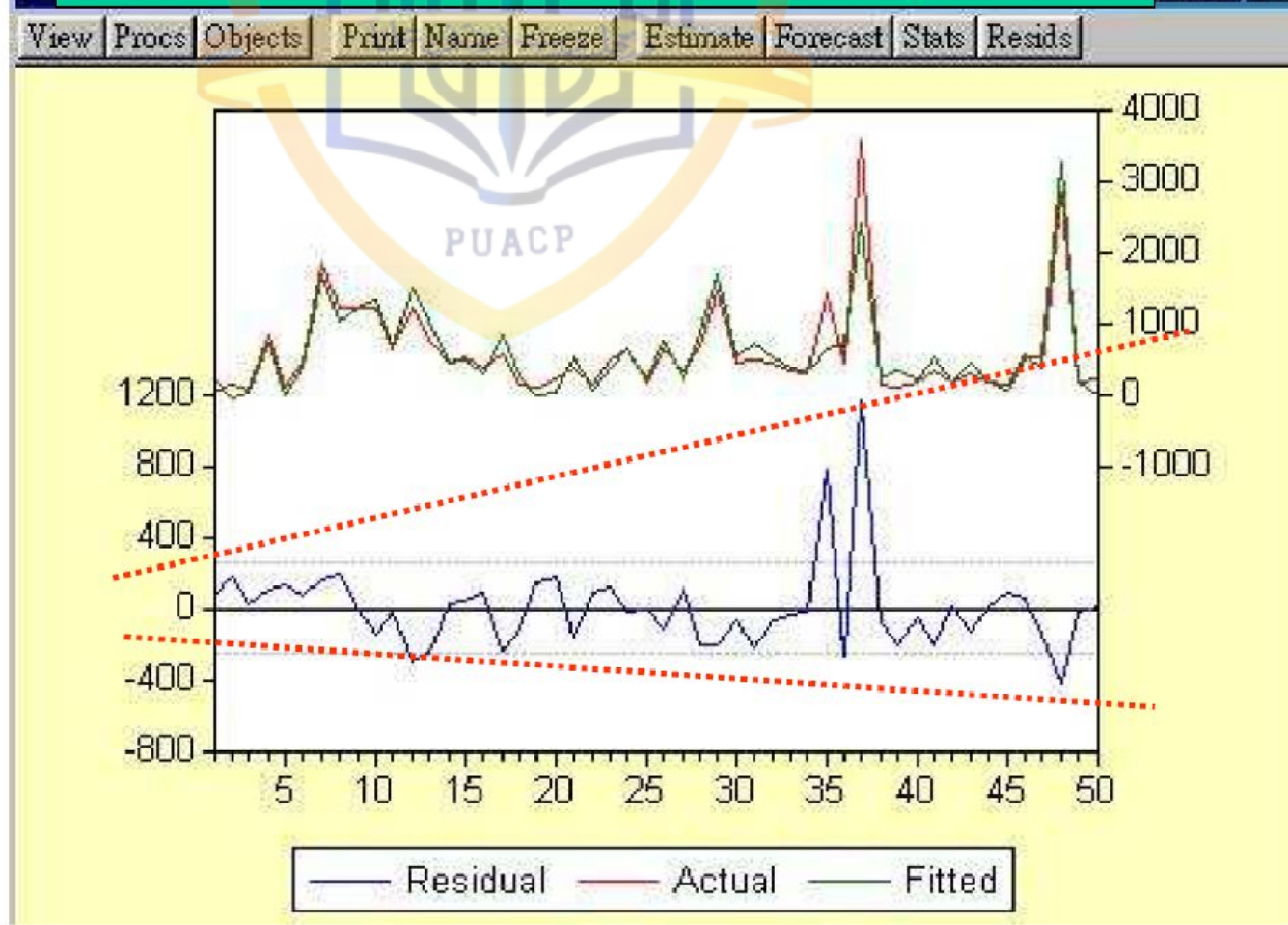
Dependent Variable: PCON
Method: Least Squares
Date: 04/03/02 Time: 10:31
Sample: 1 50
Included observations: 50

Step 1

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	551.6880	186.2709	2.961750	0.0048
REG	0.186132	0.011719	15.88302	0.0000
TAX	-53.59101	16.85588	-3.179365	0.0026
R-squared	0.866368	Mean dependent var	603.7000	
Adjusted R-squared	0.860682	S.D. dependent var	677.8267	
S.E. of regression	253.0010	Akaike info criterion	13.96279	
Sum squared resid	3008447.	Schwarz criterion	14.07751	
Log likelihood	-346.0697	F-statistic	152.3567	
Durbin-Watson stat	2.197170	Prob(F-statistic)	0.000000	

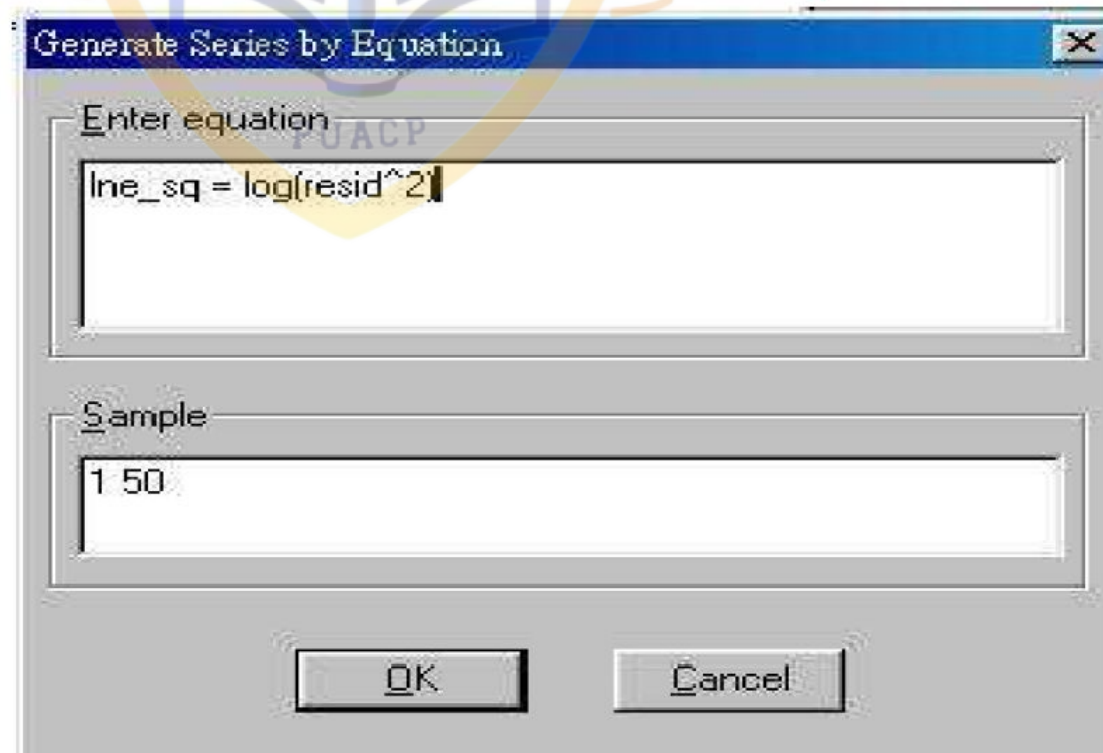
where: $PCON_i$ = petroleum consumption in the i th state (trillions of BTUs)
 TAX_i = the gasoline tax rate in the i th state (cents per gallon)
 REG_i = motor vehicle registrations in the i th state (thousands)

Check the residuals whether they are spreading out or not?



Step 2:

Obtain the residuals from previous regression,
take squares and take logs



Generate Series by Equation

Enter equation

$\text{lne_sq} = \log(\text{resid}^2)$

Sample

1 50

OK Cancel

Step 3 & 4

Equation: EQ02 Workfile: TABLE10-1

View Procs Objects Print Name Freeze

Dependent Variable: LNE_SQ

Method: Least Squares

Date: 04/03/02 Time: 10:42

Sample: 1 50

Included observations: 50

$$\ln(\hat{u}^2) = \beta_1^* + \beta_2^* \ln X_i + v_i$$

If $|t| > t^c \Rightarrow$ reject H_0
 $\Rightarrow$ heteroscedasticity

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	2.272969	2.321536	0.979080	0.3324
LOG(REG)	-0.271961	0.301431	-0.902233	0.3714
R-squared	0.016676	Mean dependent var		0.194467
Adjusted R-squared	-0.003810	S.D. dependent var		2.025391
S.E. of regression	2.029245	Akaike info criterion		4.292383
Sum squared resid	197.6561	Schwarz criterion		4.368864
Log likelihood	-105.3096	F-statistic		0.814024
Durbin-Watson stat	2.012122	Prob(F-statistic)		0.371438

Refers to Studenmund (2001), Eq.(10.26), pp.373

(ii) Glejser Test

- Glejser used the following functional forms:

$$|\hat{u}_i| = \beta_1 + \beta_2 X_i + v_i$$

Use t-test to test:

$$|\hat{u}_i| = \beta_1 + \beta_2 \sqrt{X_i} + v_i$$

$H_0 : \beta_2 = 0$ (Homoscedasticity)

$$|\hat{u}_i| = \beta_1 + \beta_2 \frac{1}{X_i} + v_i$$

$$|\hat{u}_i| = \beta_1 + \beta_2 \frac{1}{\sqrt{X_i}} + v_i$$

$$|\hat{u}_i| = \sqrt{\beta_1 + \beta_2 X_i} + v_i$$

$$|\hat{u}_i| = \sqrt{\beta_1 + \beta_2 X_i^2} + v_i$$

$$|\hat{u}_i| = \sqrt{\beta_1 + \beta_2 X_i} + v_i$$

and

$$|\hat{u}_i| = \sqrt{\beta_1 + \beta_2 X_i^2} + v_i$$

are nonlinear in the parameters and therefore cannot be estimated with the usual OLS procedure.

Glejser Test

EXAMPLE 11.2

Relationship

between

Compensation

and Productivity:

The Glejser Test

Continuing with Example 11.1, the absolute value of the residuals obtained from regression (11.5.3) were regressed on average productivity (X), giving the following results:

$$|\hat{u}_i| = 407.2783 - 0.0203X_i$$

$$se = (633.1621) \quad (0.0675) \quad r^2 = 0.0127$$

$$t = (0.6432) \quad (-0.3012)$$

(11.5.5)

As you can see from this regression, there is no relationship between the absolute value of the residuals and the regressor, average productivity. This reinforces the conclusion based on the Park test.

(iii) Spearman's Rank Correlation Test

Step 1. Fit the regression to the data on Y and X and obtain the residuals $\hat{u}_i$.

Step 2. Ignoring the sign of $\hat{u}_i$, that is, taking their absolute value $|\hat{u}_i|$, rank both $|\hat{u}_i|$ and X_i (or $\hat{Y}_i$) according to an ascending or descending order and compute the Spearman's rank correlation coefficient given previously.

Step 3. Assuming that the population rank correlation coefficient ρ_s is zero and $n > 8$, the significance of the sample r_s can be tested by the t test as follows¹⁶:

$$t = \frac{r_s \sqrt{n-2}}{\sqrt{1-r_s^2}}$$

df = n-2
Use **t-test** to test:

$H_0 : r_s = 0$ (Homoscedasticity)

$$r_s = 1 - 6 \left[\frac{\sum d_i^2}{n(n^2 - 1)} \right]$$

df = n-2

where d_i = difference in the ranks

Spearman's Rank Correlation Test

EXAMPLE 11.3

Illustration of the Rank Correlation Test

To illustrate the rank correlation test, consider the data given in Table 11.2. The data pertain to the average annual return (E_i , %) and the standard deviation of annual return (σ_i , %) of 10 mutual funds.

TABLE 11.2 Rank Correlation Test of Heteroscedasticity

Name of Mutual Fund	E_i , Average Annual Return, %	σ_i , Standard Deviation of Annual Return, %	$\hat{E}_i^\dagger$	$ \hat{u}_i ^\ddagger$ Residuals, $ (E_i - \hat{E}_i) $	Rank of $ \hat{u}_i $	Rank of σ_i	d , Difference between Two Rankings	d^2
Boston Fund	12.4	12.1	11.37	1.03	9	4	5	25
Delaware Fund	14.4	21.4	15.64	1.24	10	9	1	1
Equity Fund	14.6	18.7	14.40	0.20	4	7	-3	9
Fundamental Investors	16.0	21.7	15.78	0.22	5	10	-5	25
Investors Mutual	11.3	12.5	11.56	0.26	6	5	1	1
Loomis-Sales Mutual Fund	10.0	10.4	10.59	0.59	7	2	5	25
Massachusetts Investors Trust	16.2	20.8	15.37	0.83	8	8	0	0
New England Fund	10.4	10.2	10.50	0.10	3	1	2	4
Putnam Fund of Boston	13.1	16.0	13.16	0.06	2	6	-4	16
Wellington Fund	11.3	12.0	11.33	0.03	1	3	-2	4
Total							0	110

† Obtained from the regression: $\hat{E}_i = 5.8194 + 0.4590 \sigma_i$.

‡ Absolute value of the residuals.

Note: The ranking is in ascending order of values.

(Continued)

Spearman's Rank Correlation Test

Applying formula (11.5.6), we obtain

$$\begin{aligned} r_s &= 1 - 6 \frac{110}{10(100 - 1)} \\ &= 0.3333 \end{aligned} \quad (11.5.8)$$

Applying the t test given in Eq. (11.5.7), we obtain

$$\begin{aligned} t &= \frac{(0.3333)(\sqrt{8})}{\sqrt{1 - 0.1110}} \\ &= 0.9998 \end{aligned} \quad (11.5.9)$$

For 8 df this t value is not significant even at the 10 percent level of significance; the p value is 0.17. Thus, there is no evidence of a systematic relationship between the explanatory variable and the absolute values of the residuals, which might suggest that there is no heteroscedasticity.

(iv) The Goldfeld-Quandt Test

H_0 : homoscedasticity $\text{Var}(u_i) = \sigma^2$

H_1 : heteroscedasticity $\text{Var}(u_i) = \sigma_i^2$

Goldfeld-Quandt Test procedures:

- (1) Order or rank the observations according to the values of X_i , beginning with the lowest X value.
- (2) Omit c central observations, where c is specified a priori, and divide the remaining $(n-c)$ observation into two groups each of $(n-c)/2$ observations.
- (3) Run the separate regression on two sub-samples and obtain the respective RSS_1 and RSS_2 . Each RSS has $[(n-c)/2 - k]$ df
- (4) Compute the λ -ratio: $\lambda = \frac{RSS_2/df_2}{RSS_1/df_1}$
- (5) Compare the λ and the F^c , if $\lambda > F^c_{(\alpha, ((n-c)/2-k), ((n-c)/2-k)}$
 $\implies$ reject the H_0

Gujarati(2003)Table 11.3

obs	X	Y
1	80.0000	55.0000
2	100.0000	65.0000
3	85.0000	70.0000
4	110.0000	80.0000
5	120.0000	79.0000
6	115.0000	84.0000
7	130.0000	98.0000
8	140.0000	95.0000
9	125.0000	90.0000
10	90.0000	75.0000
11	105.0000	74.0000
12	160.0000	110.0000
13	150.0000	113.0000
14	165.0000	125.0000
15	145.0000	108.0000
16	180.0000	115.0000
17	225.0000	140.0000
18	200.0000	120.0000
19	240.0000	145.0000
20	185.0000	130.0000
21	220.0000	152.0000
22	210.0000	144.0000
23	245.0000	175.0000
24	260.0000	180.0000
25	190.0000	135.0000
26	205.0000	140.0000
27	265.0000	178.0000
28	270.0000	191.0000
29	230.0000	137.0000
30	250.0000	189.0000

Re-order data

RX	RY
80.0000	55.0000
85.0000	70.0000
90.0000	75.0000
100.0000	65.0000
105.0000	74.0000
110.0000	80.0000
115.0000	84.0000
120.0000	79.0000
125.0000	90.0000
130.0000	98.0000
140.0000	95.0000
145.0000	108.0000
150.0000	113.0000
160.0000	110.0000
165.0000	125.0000
180.0000	115.0000
185.0000	130.0000
190.0000	135.0000
200.0000	120.0000
205.0000	140.0000
210.0000	144.0000
220.0000	152.0000
225.0000	140.0000
230.0000	137.0000
240.0000	145.0000
245.0000	175.0000
250.0000	189.0000
260.0000	180.0000
265.0000	178.0000
270.0000	191.0000

Run regression based
on this group of data ($n_1=13$),
Obtains RSS_1

Omit central observations
($c = 4$)

Run regression based
on this group of data ($n_2=13$),
Obtains RSS_2

$$\lambda = \frac{RSS_2 / df_2}{RSS_1 / df_1} > F^c ?$$

Continued..

Regression based on the first 13 observations:

$$\hat{Y}_i = 3.4094 + 0.6968X_i$$

(8.7049) (0.0744) $r^2 = 0.8887$ $RSS_1 = 377.17$ $df = 11$

Regression based on the last 13 observations:

$$\hat{Y}_i = -28.0272 + 0.7941X_i$$

(30.6421) (0.1319) $r^2 = 0.7681$ $RSS_2 = 1536.8$ $df = 11$

From these results we obtain

$$\lambda = \frac{RSS_2/df}{RSS_1/df} = \frac{1536.8/11}{377.17/11}$$
$$\lambda = 4.07$$

The critical F value for 11 numerator and 11 denominator df at the 5 percent level is 2.82. Since the estimated $F (= \lambda)$ value exceeds the critical value, we may conclude that there is heteroscedasticity in the error variance. However, if the level of significance is fixed at 1 percent, we may not reject the assumption of homoscedasticity. (Why?) Note that the p value of the observed λ is 0.014.

(v) Breusch–Pagan–Godfrey Test

$$Y_i = \beta_1 + \beta_2 X_{2i} + \cdots + \beta_k X_{ki} + u_i$$

Assume that the error variance σ_i^2 is described as

$$\sigma_i^2 = \alpha_1 + \alpha_2 Z_{2i} + \cdots + \alpha_m Z_{mi}$$

some or all of the X's can serve as Z's

Step 1. Estimate by OLS and obtain the residuals $\hat{u}_1, \hat{u}_2, \dots, \hat{u}_n$.

Step 2. Obtain $\tilde{\sigma}^2 = \sum \hat{u}_i^2 / n$. Recall from Chapter 4 that this is the maximum likelihood (ML) estimator of σ^2 . [Note: The OLS estimator is $\sum \hat{u}_i^2 / (n - k)$.]

Step 3. Construct variables p_i defined as

$$p_i = \hat{u}_i^2 / \tilde{\sigma}^2$$

which is simply each residual squared divided by $\tilde{\sigma}^2$.

Step 4. Regress p_i thus constructed on the Z's as

$$p_i = \alpha_1 + \alpha_2 Z_{2i} + \cdots + \alpha_m Z_{mi} + v_i$$

(v) Breusch–Pagan–Godfrey Test

Step 5. Obtain the ESS (explained sum of squares) from

$$\Theta = \frac{1}{2}(\text{ESS})$$

$$\Theta \underset{\text{asy}}{\sim} \chi_{m-1}^2$$

that is, Θ follows the chi-square distribution with $(m - 1)$ degrees of freedom. (*Note: asy* means asymptotically.)

Therefore, if in an application the computed $\Theta (= \chi^2)$ exceeds the critical χ^2 value at the chosen level of significance, one can reject the hypothesis of homoscedasticity; otherwise one does not reject it.

Breusch–Pagan–Godfrey Test

EXAMPLE 11.5 The Breusch– Pagan–Godfrey (BPG) Test

As an example, let us revisit the data (Table 11.3) that were used to illustrate the Goldfeld–Quandt heteroscedasticity test. Regressing Y on X , we obtain the following:

Step 1.

$$\begin{aligned}\hat{Y}_i &= 9.2903 + 0.6378X_i \\ \text{se} &= (5.2314) \quad (0.0286) \quad \text{RSS} = 2361.153 \quad R^2 = 0.9466\end{aligned}\quad (11.5.18)$$

Step 2.

$$\hat{\sigma}^2 = \sum \hat{u}_i^2 / 30 = 2361.153 / 30 = 78.7051$$

Step 3. Divide the squared residuals $\hat{u}_i$ obtained from regression (11.5.18) by 78.7051 to construct the variable p_i .

Step 4. Assuming that p_i are linearly related to $X_i (= Z_i)$ as per Eq. (11.5.14), we obtain the regression

$$\begin{aligned}\hat{p}_i &= -0.7426 + 0.0101X_i \\ \text{se} &= (0.7529) \quad (0.0041) \quad \text{ESS} = 10.4280 \quad R^2 = 0.18\end{aligned}\quad (11.5.19)$$

Step 5.

$$\Theta = \frac{1}{2}(\text{ESS}) = 5.2140\quad (11.5.20)$$

Under the assumptions of the BPG test Θ in Eq. (11.5.20) asymptotically follows the chi-square distribution with 1 df. (Note: There is only one regressor in Eq. [11.5.19].) Now from the chi-square table we find that for 1 df the 5 percent critical chi-square value is 3.8414 and the 1 percent critical χ^2 value is 6.6349. Thus, the observed chi-square value of 5.2140 is significant at the 5 percent but not the 1 percent level of significance. Therefore, we reach the same conclusion as the Goldfeld–Quandt test.

(vi) White's heteroscedasticity test (no cross terms)
(LM test)

H_0 : homoscedasticity $\text{Var} (u_i) = \sigma^2$

H_1 : heteroscedasticity $\text{Var} (u_i) = \sigma_i^2$

Test procedures:

(1) Run OLS on regression: $Y_i = \beta_1 + \beta_2 X_{2i} + \beta_3 X_{3i} + \dots + \beta_q X_{qi} + u_i$,
 obtain the residuals, $\hat{u}_i$

(2) Run the auxiliary regression:

$$\hat{u}_i^2 = \delta_1 + \delta_2 X_{2i} + \dots + \delta_q X_{qi} + \delta_{q+1} X_{2i}^2 + \dots + \delta_{2q-1} X_{qi}^2 + v_i$$

$$H_0: \delta_2 = \delta_3 = \dots = \delta_{2q-1} = 0$$

(3) Compute **W (or LM)** = $n \bullet R^2 \underset{asy}{\sim} \chi^2_{df}$

(4) Compare the **W** and $\chi^2_{df(=2q-1)}$ (where the **df** is # of regressors in (2))
 if **W** > $\chi^2_{df} \implies$ reject the H_0

$$Y_i = \beta_1 + \beta_2 X_{2i} + \beta_3 X_{3i} + \beta_4 X_{4i} + u_i$$

Dependent Variable: PRICE1

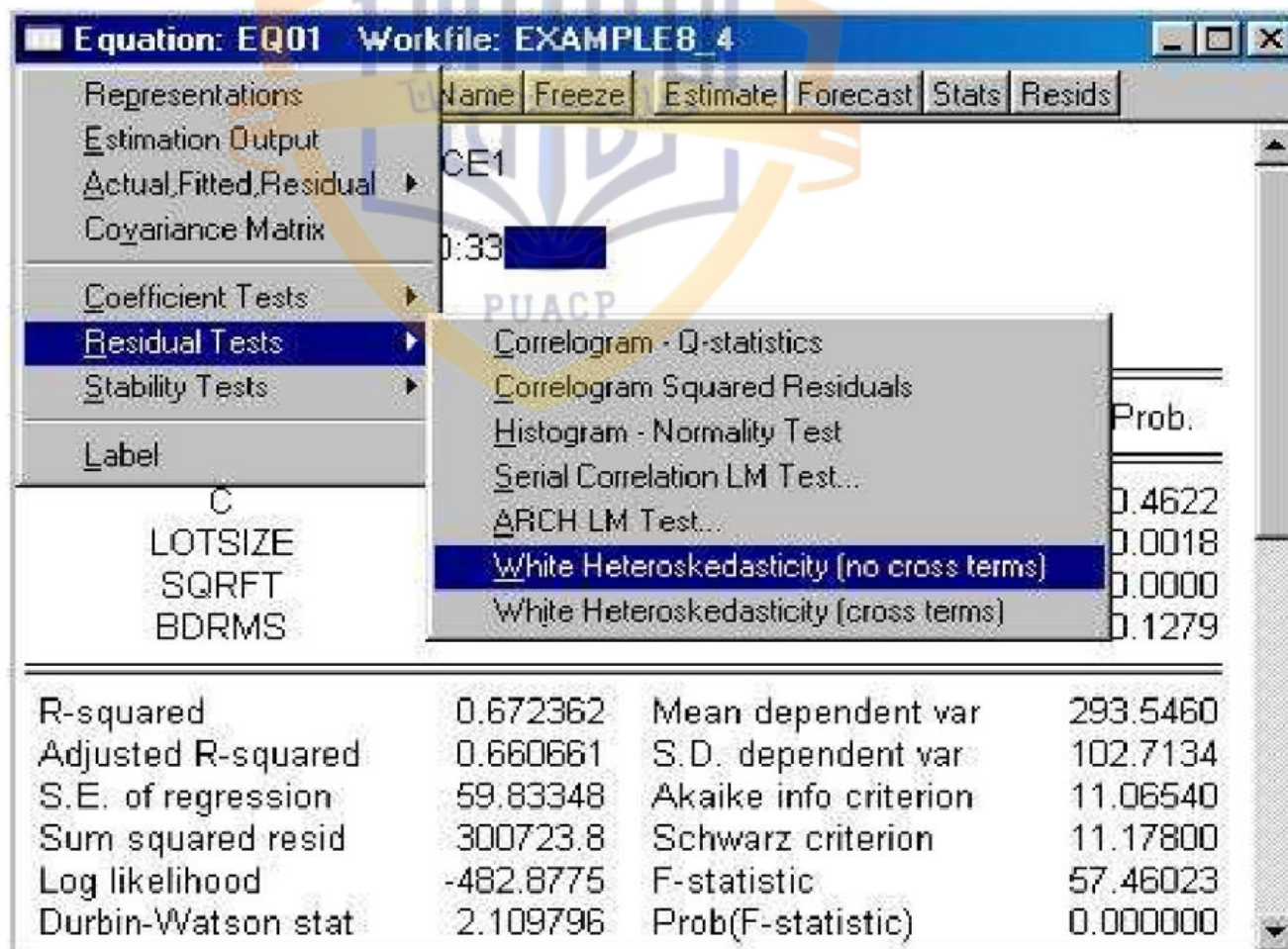
Method: Least Squares

Date: 03/27/01 Time: 10:33

Sample: 1 88

Included observations: 88

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	-21.77031	29.47504	-0.738601	0.4622
LOTSIZE	0.002068	0.000642	3.220096	0.0018
SQRFT	0.122778	0.013237	9.275093	0.0000
BDRMS	13.85252	9.010145	1.537436	0.1279
R-squared	0.672362	Mean dependent var	293.5460	
Adjusted R-squared	0.660661	S.D. dependent var	102.7134	
S.E. of regression	59.83348	Akaike info criterion	11.06540	
Sum squared resid	300723.8	Schwarz criterion	11.17800	
Log likelihood	-482.8775	F-statistic	57.46023	
Durbin-Watson stat	2.109796	Prob(F-statistic)	0.000000	



White Heteroskedasticity Test:

W =

F-statistic	4.126676	Probability	0.001151
Obs*R-squared	20.60215	Probability	0.002162

Test Equation:

Dependent Variable: RESID^2

Method: Least Squares

Date: 03/26/01 Time: 15:03

Sample: 1 88

Included observations: 88

W >

$$\chi^2_{(0.05, 6)} = 12.59$$

$$\chi^2_{(0.10, 6)} = 10.64$$

→ reject H_0

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	-2368.728	10999.82	-0.215342	0.8300
LOTSIZE	0.627173	0.249317	2.515565	0.0139
LOTSIZE^2	-5.02E-06	2.72E-06	-1.845148	0.0687
SQRFT	-14.0249			
SQRFT^2	0.00297			
BDRMS	7454.81			
BDRMS^2	-770.654			
R-squared	0.23411			
Adjusted R-squared	0.17738			
S.E. of regression	6434.48			
Sum squared resid	3.35E+09	Schwarz criterion	20.64999	
Log likelihood	-892.9290	F-statistic	4.126676	

The White test for a linear model
The test statistic (nR^2) indicates
heteroscedasticity is existed.

White Heteroskedasticity Test:

F-statistic	1.073611	Probability	0.385232
Obs*R-squared	6.482799	Probability	0.371331

Test Equation:

Dependent Variable: RESIDUALS
 Method: Least Squares
 Date: 03/26/01 Time: 15:51
 Sample: 1 88
 Included observations: 88

$W < \chi^2_{(0.05, 6)} = 12.59$
 $\chi^2_{(0.10, 6)} = 10.64$ → not reject H_0

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	6.485362	5.704938	1.136798	0.2590
LOG(LOTSIZE)	-0.148307	0.185090	-0.801273	0.4253
(LOG(LOTSIZE))^2	0.007679	0.010029	0.765679	0.4461
LOG(SQRFT)	-1.471960	1.501275	-0.980474	0.3298
(LOG(SQRFT))^2	0.091800	0.098166	0.935153	0.3525
LOG(BDRMS)	0.148320	0.260374	0.569644	0.5705

After transform the linear model to a log-log model.

The White test for a log-log model of the same data

The W-statistic (nR^2) indicates heteroscedasticity is not existed

(vii) White's general heteroscedasticity test (with cross terms)

H_0 : homoscedasticity $\text{Var}(u_i) = \sigma^2$

H_1 : heteroscedasticity $\text{Var}(u_i) = \sigma_i^2$

Test procedures:

(1) Run OLS on regression: $Y_i = \beta_1 + \beta_2 X_{2i} + \beta_3 X_{3i} + u_i$,
obtain the residuals, $\hat{u}_i$

(2) Run the auxiliary regression:

$$\hat{u}_i^2 = \alpha_1 + \alpha_2 X_{2i} + \alpha_3 X_{3i} + \alpha_4 X_{2i}^2 + \alpha_5 X_{3i}^2 + \alpha_6 X_{2i} X_{3i} + v_i$$

(3) Compute **W (or LM)** = $n \bullet R^2 \underset{asy}{\sim} \chi^2_{df}$

(4) Compare the **W** and χ^2_{df} (where the **df** is # of regressors in (2))
if **W** > $\chi^2_{df} \implies$ reject the H_0

Equation: UNTITLED Workfile: TAB0101

Representations Freeze Estimate Forecast Stats Resids

Estimation Output

Actual, Fitted, Residual

Gradients and Derivatives

Covariance Matrix

Coefficient Tests

Residual Tests

Stability Tests

Label

R-squared	0.600	Correlogram - Q-statistics	0.0000
Adjusted R-squared	0.541	Correlogram Squared Residuals	0.0341
S.E. of regression	1.284183	Histogram - Normality Test	0.0000
Sum squared resid	9850.106	Serial Correlation LM Test...	0.0000
Log likelihood	-57.28925	ARCH LM Test...	0.0000
Durbin-Watson stat	1.284183	White Heteroskedasticity (no cross terms)	0.0000
		White Heteroskedasticity (cross terms)	0.0000
		Schwarz criterion	9.962359
		F-statistic	1094.160
		Prob(F-statistic)	0.000000

Example 8.4 (Wooldridge, pp.258)

White Heteroskedasticity Test:

W =

F-statistic	5.386953	Probability	0.000010
Obs*R-squared	33.73166	Probability	0.000100

Test Equation:

Dependent Variable: RESID^2

Method: Least Squares

Date: 03/26/01 Time: 15:09

Sample: 1 88

Included observations: 88

W >

$$\chi^2_{(0.05, 9)} = 16.92$$

$$\chi^2_{(0.10, 9)} = 14.68$$

→ reject H_0

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	15626.24	11369.41	1.374411	0.1733
LOTSIZE	-1.859507	0.637097	-2.918719	0.0046
LOTSIZE^2	-4.98E-05	0.000000	-0.000000	0.9999
LOTSIZE*SQRFT	0.00045	0.000000	0.000000	0.9999
LOTSIZE*BDRMS	0.31464	0.000000	0.000000	0.9999
SQRFT	-2.67391	0.000000	0.000000	0.9999
SQRFT^2	0.00035	0.000000	0.000000	0.9999
SQRFT*BDRMS	-1.02086	0.000000	0.000000	0.9999
BDRMS	-1982.84	0.000000	0.000000	0.9999
BDRMS^2	289.754	0.000000	0.000000	0.9999

The White test for a linear model
The test statistic indicates
heteroscedasticity is existed.

R-squared	0.383314	Mean dependent var	3417.316
Adjusted R-squared	0.312158	S.D. dependent var	7094.384
S.E. of regression	5883.814	Akaike info criterion	20.30444

White Heteroskedasticity Test: **For the log-log model**

F-statistic	1.152979	Probability	0.336903
Obs*R-squared	10.33257	Probability	0.324239

Test Equation:

Dependent Variable: RESID^2

Method: Least Squares

Date: 03/26/01 Time: 15:18

Sample: 1 88

Included observations: 88

W <

$$\chi^2_{(0.05, 9)} = 16.92$$

$$\chi^2_{(0.10, 9)} = 14.68$$

→ not reject H_0

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	12.32744	6.568291	1.876811	0.0643
LOG(LOTSIZE)	-1.154355			
(LOG(LOTSIZE))^2	0.020120			
(LOG(LOTSIZE))*(LOG	0.113041			
(LOG(LOTSIZE))*(LOG	-0.067503			
LOG(SQRFT)	-2.007544			
(LOG(SQRFT))^2	0.067256			
(LOG(SQRFT))*(LOG	-0.079257			
LOG(BDRMS)	1.302747			
(LOG(BDRMS))^2	-0.016190	0.152561	-0.106124	0.9158
R-squared	0.117416	Mean dependent var		0.032857
Adjusted R-squared	0.015579	S.D. dependent var		0.073045

The White test for a log-log model
The test statistic indicates
heteroscedasticity is not existed

11.6 REMEDIAL MEASURES

- As we have seen, heteroscedasticity does not destroy the unbiasedness and consistency properties of the OLS estimators, but they are no longer efficient, not even asymptotically (i.e., large sample size). This lack of efficiency makes the usual hypothesis-testing procedure of dubious value. Therefore, remedial measures may be called for. There are two approaches to remediation:
 - when σ_i^2 is known
 - when σ_i^2 is not known.
 - when σ_i^2 is Known: We use the Method of Weighted Least Squares
 - Example 11.7

11.6 REMEDIAL MEASURES

- **When σ_i^2 Is Not Known**

- As noted earlier, if true σ_i^2 are known, we can use the WLS method to obtain BLUE estimators.
- Since the true σ_i^2 are rarely known, is there a way of obtaining *consistent* (in the statistical sense) estimates of the variances and covariances of OLS estimators even if there is heteroscedasticity? The answer is yes.

- **White's Heteroscedasticity-Consistent Variances and Standard Errors**

- White has shown that this estimate can be performed so that *asymptotically* valid (i.e., large-sample) statistical inferences can be made about the true parameter values.
- White's heteroscedasticity-corrected standard errors are also known as robust standard errors.
- Example 11.8

11.6 REMEDIAL MEASURES

- As the preceding results show, (White's) heteroscedasticity-corrected standard errors are considerably larger than the OLS standard errors and therefore the estimated t values are much smaller than those obtained by OLS. On the basis of the latter, both the regressors are statistically significant at the 5 percent level, whereas on the basis of White estimators they are not.
- However, it should be pointed out that White's heteroscedasticity-corrected standard errors can be larger or smaller than the uncorrected standard errors. Since White's heteroscedasticity-consistent estimators of the variances are now available in established regression packages, it is recommended that the reader report them. As Wallace and Silver note:

11.6 REMEDIAL MEASURES

- Generally speaking, it is probably a good idea to use the WHITE option [available in regression programs] routinely, perhaps comparing the output with regular OLS output as a check to see whether heteroscedasticity is a serious problem in a particular set of data.
- Please read Example 9, 10, 11 from textbook